

IMAT REPAIR DRILL — Repeat Offenders

70 questions · 100 minutes · scoring: +1.5 correct, −0.4 wrong, 0 blank · Repair drill · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Respiration & Bioenergetics (Q1-12)

1. In a eukaryotic cell, the enzymes that catalyse the citric acid (Krebs) cycle are located mainly in the:

- A) intermembrane space
- B) cytosol
- C) mitochondrial matrix
- D) inner mitochondrial membrane
- E) outer mitochondrial membrane

Answer: C

The Krebs cycle enzymes sit free in the mitochondrial matrix, where acetyl-CoA enters the cycle (succinate dehydrogenase is the one exception, bound to the inner membrane). Placing the cycle in the inner membrane confuses it with the electron transport chain and ATP synthase, which are oxidative phosphorylation machinery, and glycolysis rather than the cycle runs in the cytosol.

2. Which of the following processes occurs in the mitochondrial matrix?

- A) reduction of O₂ to water at complex IV
- B) electron transfer between respiratory complexes I and IV
- C) conversion of glucose to pyruvate
- D) ATP synthesis driven by a proton gradient
- E) oxidation of acetyl-CoA to CO₂ in the citric acid cycle

Answer: E

Oxidising acetyl-CoA to CO₂ in the citric acid cycle happens in the matrix, where the cycle's enzymes sit. ATP synthesis driven by the proton gradient, electron transfer between complexes I and IV, and the reduction of O₂ to water all occur at the inner mitochondrial membrane, while conversion of glucose to pyruvate is glycolysis and happens in the cytosol. The recurring error is putting the whole respiratory chain inside the matrix.

3. In human cells, ATP synthase is located in the:

- A) outer mitochondrial membrane
- B) intermembrane space
- C) cytosol
- D) inner mitochondrial membrane
- E) mitochondrial matrix

Answer: D

ATP synthase is embedded in the inner mitochondrial membrane, with its catalytic head facing the matrix. It is driven by the H⁺ gradient between the intermembrane space and the matrix that the electron transport chain builds. Saying it sits free in the matrix confuses the site of the Krebs cycle with the site of oxidative phosphorylation.

4. Cyanide blocks complex IV, the last complex of the electron transport chain. What is the most immediate effect on ATP synthase?

- A) It stops making ATP because the H^+ gradient across the inner membrane dissipates.
- B) It continues making ATP because it phosphorylates ADP by substrate-level phosphorylation.
- C) It makes more ATP because protons accumulate in the matrix.
- D) It keeps working normally because ATP synthase is not part of the electron transport chain.
- E) It switches to using electrons taken directly from glucose.

Answer: A

Step 1: cyanide blocks complex IV, so electrons cannot pass to O_2 and the chain backs up. Step 2: proton pumping into the intermembrane space stops, so the H^+ gradient across the inner membrane dissipates. Step 3: ATP synthase is not itself an electron transport complex, but it depends entirely on that gradient, so ATP synthesis halts. The claim that it keeps working normally ignores this dependency, and protons would not accumulate in the matrix to boost output, because the gradient runs the other way.

5. During the complete aerobic oxidation of one glucose molecule, carbon dioxide is released:

- A) during glycolysis only
- B) during pyruvate oxidation and the citric acid cycle
- C) during glycolysis and the electron transport chain
- D) during substrate-level phosphorylation only
- E) during the electron transport chain only

Answer: B

CO_2 leaves whenever carbon exits the fuel: two molecules are released in pyruvate oxidation (the link reaction) and four more in the citric acid cycle, six per glucose. Glycolysis splits glucose into two pyruvates with no carbon lost, and the electron transport chain shuttles electrons, not carbon, so it releases none. Locating CO_2 release in the electron transport chain is therefore impossible.

6. A working muscle cell temporarily carries out only glycolysis followed by lactic acid fermentation. How many molecules of CO_2 are released per molecule of glucose consumed?

- A) 0 B) 1 C) 2 D) 3 E) 6

Answer: A

Glycolysis converts glucose to two pyruvates with no carbon lost, and lactic acid fermentation then reduces pyruvate straight to lactate with no decarboxylation step. So zero CO_2 molecules are released. The tempting count of two wrongly imports the CO_2 output of pyruvate oxidation, a step that does not occur in this anaerobic pathway.

7. Most of the carbon dioxide carried in human blood is transported:

- A) as carbonic acid bound to albumin
- B) dissolved as CO_2 in the plasma
- C) as carbaminohemoglobin bound to hemoglobin
- D) attached to the heme iron of hemoglobin
- E) as bicarbonate ions (HCO_3^-) in the plasma

Answer: E

About 70% of blood CO_2 travels as bicarbonate ions: carbonic anhydrase in red blood cells converts CO_2 and water to carbonic acid, which dissociates to HCO_3^- that then diffuses into the plasma. Carbaminohemoglobin accounts for only about 23% and about 7% stays dissolved. Also, CO_2 binds the globin protein part of hemoglobin, while the heme iron is reserved for O_2 .

8. Approximately what percentage of the carbon dioxide in human blood is transported as bicarbonate ions (HCO_3^-)?

- A) about 23% B) about 7% C) about 97% D) about 70% E) about 50%

Answer: D

Roughly 70% of CO_2 is carried as bicarbonate, formed by carbonic anhydrase in red blood cells. The 23% figure belongs to carbaminohemoglobin and 7% to CO_2 dissolved in plasma, so those two answers come from swapping the genuine transport pools, and 97% is the familiar O_2 saturation of hemoglobin, not a CO_2 statistic.

9. When a cell's ATP concentration is high, the rate of glycolysis falls. This happens because ATP:

- A) competes with fructose-6-phosphate for the active site of phosphofructokinase-1
B) activates phosphofructokinase-1 and accelerates glucose breakdown
C) binds allosterically to phosphofructokinase-1 and lowers its activity
D) competitively inhibits pyruvate kinase
E) irreversibly blocks the active site of hexokinase

Answer: C

High ATP binds an allosteric regulatory site on phosphofructokinase-1, the rate-limiting enzyme of glycolysis, and lowers its activity. It is not competitive inhibition: ATP binds away from the active site, so it does not compete with fructose-6-phosphate for the substrate pocket. Saying ATP activates PFK-1 gets the direction of regulation backwards, since AMP is the activator.

10. In a eukaryotic cell, complete oxidation of one glucose molecule yields roughly 30-32 ATP. Most of this ATP is produced by:

- A) lactic acid fermentation inside the mitochondria
B) oxidative phosphorylation powered by the electron transport chain
C) substrate-level phosphorylation in the citric acid cycle
D) direct phosphorylation during pyruvate oxidation
E) substrate-level phosphorylation in glycolysis

Answer: B

Substrate-level phosphorylation delivers only 2 ATP from glycolysis plus 2 GTP from the citric acid cycle, about 4 of the 30-32 total. The remaining roughly 26-28 ATP come from oxidative phosphorylation: NADH and FADH_2 supply electrons to the transport chain, which pumps protons to power ATP synthase. Picking any substrate-level step mistakes the small direct yield for the bulk of the total.

11. During skeletal muscle contraction, the myosin heads attach directly to:

- A) calcium ions B) the Z disc C) actin D) troponin E) tropomyosin

Answer: C

In cross-bridge cycling the myosin heads bind directly to actin and pivot to pull the thin filaments. Calcium binds troponin, which shifts tropomyosin off the myosin-binding sites on actin, so the regulatory proteins act only indirectly. Myosin never attaches to troponin, tropomyosin or calcium during contraction.

12. When oxygen is unavailable, yeast and muscle cells run fermentation chiefly in order to:

- A)** produce NADH for the electron transport chain
- B)** convert NAD⁺ into NADP⁺
- C)** remove excess NAD⁺ from the cytosol
- D)** generate extra ATP beyond that made in glycolysis
- E)** regenerate NAD⁺ so that glycolysis can continue

Answer: E

Fermentation's job is to oxidise the NADH from glycolysis back to NAD⁺ (pyruvate to lactate in muscle, to ethanol and CO₂ in yeast). Without recycled NAD⁺, glycolysis would stall once its supply ran out and the net 2 ATP per glucose would stop. The fermentation steps themselves generate no ATP, so extra ATP beyond glycolysis is a mirage.

Section B - Body Systems (Q13-20)

13. Bile, which is produced by the liver and stored in the gallbladder, assists fat digestion primarily by...

- A)** emulsifying large fat globules into tiny droplets, increasing the surface area available to lipase
- B)** hydrolysing triglycerides into fatty acids and glycerol
- C)** lowering the pH of chyme so that pancreatic enzymes can act
- D)** secreting lipase, the enzyme that chemically digests fats
- E)** activating pepsinogen into pepsin in the stomach

Answer: A

Bile contains no enzymes. Its bile salts emulsify large fat globules into tiny droplets, greatly increasing the surface area that pancreatic lipase can attack. Hydrolysing triglycerides is lipase's chemical job, and converting pepsinogen to pepsin is what stomach acid does, not bile.

14. During a normal inhalation, which sequence of events occurs?

- A)** The diaphragm relaxes and domes upward, thoracic volume decreases, and pressure inside the lungs rises above atmospheric.
- B)** The diaphragm contracts and flattens, thoracic volume increases, and pressure inside the lungs rises above atmospheric.
- C)** The diaphragm relaxes and domes upward, thoracic volume increases, and pressure in the lungs falls below atmospheric.
- D)** The diaphragm contracts and flattens, thoracic volume increases, and pressure in the lungs falls below atmospheric, drawing air in.
- E)** The diaphragm contracts and flattens, thoracic volume decreases, and pressure inside the lungs rises.

Answer: D

Step 1: the diaphragm contracts and flattens. Step 2: the thoracic cavity enlarges, so thoracic volume increases. Step 3: the larger volume lowers the pressure inside the lungs below atmospheric, and air flows in down that pressure difference. Accounts with decreasing volume or rising lung pressure describe exhalation and contradict inhalation mechanics.

15. During quiet breathing at rest, expiration is best described as...

- A)** a passive process in which the diaphragm contracts while thoracic volume stays constant
- B)** a passive process in which the diaphragm relaxes and domes upward, reducing thoracic volume
- C)** an active process in which the abdominal muscles contract forcibly to empty the lungs
- D)** an active process in which the diaphragm contracts and flattens to push air out
- E)** a passive process in which the diaphragm relaxes while thoracic volume increases

Answer: B

Quiet expiration is passive: the diaphragm relaxes and domes upward, thoracic volume falls, and the lungs' elastic recoil pushes air out. Contracting and flattening is the inhalation movement, and forcible abdominal muscle contraction belongs to forced expiration, not quiet breathing.

16. Oxygenated blood is delivered to the systemic circulation by which route?

- A)** From the left ventricle into the pulmonary artery
- B)** From the left ventricle into the aorta
- C)** From the right atrium into the vena cava
- D)** From the left atrium into the pulmonary vein
- E)** From the right ventricle into the aorta

Answer: B

Oxygenated blood returns from the lungs in the pulmonary veins to the left atrium, moves to the left ventricle, and is pumped into the aorta to supply the body. The right ventricle instead sends deoxygenated blood through the pulmonary artery to the lungs. Any route through the right side of the heart or the pulmonary artery describes pulmonary, not systemic, delivery.

17. The semilunar valves of the heart prevent...

- A)** blood from flowing back from the atria into the venae cavae and pulmonary veins
- B)** blood from flowing back from the ventricles into the atria
- C)** blood from flowing from the atria into the ventricles during diastole
- D)** blood from flowing back from the aorta and pulmonary artery into the ventricles
- E)** blood from flowing from the pulmonary veins into the left atrium

Answer: D

The semilunar valves guard the ventricular outlets in the aorta and pulmonary artery and snap shut as the ventricles relax, stopping ejected blood from falling back into them. Preventing backflow from ventricles into atria is the job of the atrioventricular valves (tricuspid and mitral), a different valve set.

18. The urinary bladder must tolerate repeated stretching as it fills with urine. Which epithelium lines it?

- A)** Stratified squamous epithelium
- B)** Simple squamous epithelium
- C)** Simple columnar epithelium
- D)** Ciliated pseudostratified epithelium
- E)** Transitional epithelium

Answer: E

Transitional epithelium (urothelium) lines the bladder and ureters: its rounded surface cells flatten as the organ fills, allowing repeated stretching and recoil without leakage. Simple columnar epithelium is built for secretion and absorption and cannot survive that distension, and stratified squamous is the abrasion-resistant lining of skin and oesophagus, not a stretchy one.

19. The kidney tubules and many secretory glands are lined mainly by which epithelium?

- A) Simple columnar epithelium
- B) Simple squamous epithelium
- C) Simple cuboidal epithelium
- D) Stratified squamous epithelium
- E) Transitional epithelium

Answer: C

Simple cuboidal epithelium lines kidney tubules and forms much of secretory glands, since its cells carry enough cytoplasm and organelles for active secretion and absorption. Simple squamous cells are paper-thin, specialised for diffusion in alveoli and vessel linings, not secretory work. Simple columnar instead dominates absorptive surfaces like the gut lining.

20. Ciliated epithelium lining the trachea serves mainly to...

- A) sweep mucus with trapped dust and microbes upward toward the pharynx
- B) contract rhythmically to narrow the airway during coughing
- C) secrete the mucus that traps inhaled particles
- D) absorb oxygen directly from the air passing through the trachea
- E) provide the main surface for gas exchange with the blood

Answer: A

The coordinated beating of the cilia sweeps the mucus blanket, with its trapped dust and microbes, upward toward the pharynx to be swallowed or expelled. The mucus is secreted by goblet cells rather than made by the cilia, and gas exchange occurs in the thin-walled alveoli, not the trachea.

Section C - Cell Division & Cell Biology (Q21-30)

21. Nondisjunction in meiosis I, meaning the failure of homologous chromosomes to separate from each other, occurs during:

- A) telophase I B) prophase I C) anaphase I D) metaphase I E) anaphase II

Answer: C

Homologous chromosomes are physically pulled to opposite poles in anaphase I, so a failure of that separation is anaphase I nondisjunction. By telophase I the chromosomes have already reached the poles, so blaming telophase comes too late. Anaphase II separates sister chromatids instead, which is the meiosis-II version of the error.

22. In a human primary spermatocyte ($2n = 46$) undergoing meiosis II, nondisjunction would be the failure of which structures to separate?

- A) chromosomes from the spindle poles
- B) bivalents at the metaphase plate
- C) homologous chromosomes of each pair
- D) sister chromatids of each chromosome
- E) non-sister chromatids at the chiasmata

Answer: D

Meiosis II separates sister chromatids, the identical copies of each chromosome joined at the centromere, so their failure to part defines meiosis-II nondisjunction. Homologous chromosomes and bivalents already separated in anaphase I, so they cannot be the structures failing in this division.

23. Independent assortment of maternal and paternal chromosomes results from their random orientation at the metaphase plate. This alignment of homologous pairs occurs during:

- A)** prophase I of meiosis
- B)** metaphase I of meiosis
- C)** metaphase of mitosis
- D)** metaphase II of meiosis
- E)** telophase I of meiosis

Answer: B

At metaphase I each homologous pair (tetrad) orients randomly relative to the spindle poles, and that orientation alone decides which maternal or paternal copy reaches each gamete, giving 2^{23} combinations in humans. By metaphase II the homologues have already separated, and only individual chromosomes align. Prophase I is for pairing and crossing over, not this alignment.

24. Crossing over, the exchange of corresponding segments between homologous chromosomes, takes place during:

- A)** prophase I **B)** anaphase I **C)** metaphase I **D)** prophase II **E)** telophase II

Answer: A

Crossing over occurs in prophase I while homologues are paired as bivalents, when chiasmata form and non-sister chromatids exchange corresponding segments. Metaphase I merely aligns the already-exchanged pairs on the spindle, so it displays the results of crossing over rather than performing it.

25. Glucose and galactose differ in the spatial arrangement of the hydroxyl group at a single chiral carbon, C-4. They are therefore classified as:

- A)** cis-trans isomers
- B)** structural isomers
- C)** enantiomers
- D)** anomers
- E)** epimers

Answer: E

Sugars differing in configuration at exactly one of several chiral centres are epimers, so glucose and galactose are C-4 epimers. Enantiomers must be mirror images at every chiral centre, which these are not, and anomers differ at the anomeric carbon (C-1) of the cyclic form, not C-4.

26. Tay-Sachs disease is caused by a defective enzyme, hexosaminidase A, so an undegraded membrane lipid (GM2 ganglioside) accumulates. In which organelle does the lipid build up?

- A)** smooth endoplasmic reticulum
- B)** peroxisome
- C)** lysosome
- D)** Golgi apparatus
- E)** mitochondrion

Answer: C

Hexosaminidase A normally degrades GM2 ganglioside inside lysosomes, the organelle holding the cell's hydrolytic enzymes, so the undigested lipid accumulates there: Tay-Sachs is a classic lysosomal storage disease. Peroxisomes instead handle very-long-chain fatty acid oxidation and catalase reactions, not ganglioside breakdown.

27. The accumulation of misfolded proteins inside the endoplasmic reticulum triggers the unfolded protein response. This misfolded-protein build-up is sensed in the:

- A)** smooth endoplasmic reticulum
- B)** cytosol
- C)** Golgi apparatus
- D)** rough endoplasmic reticulum
- E)** lysosome

Answer: D

Secretory and membrane proteins are made and folded inside the rough ER, so misfolded chains accumulate and are sensed there, triggering the unfolded protein response that slows translation and expands ER folding capacity. The smooth ER specialises in lipid synthesis and detoxification and does not fold proteins, and the Golgi only processes proteins that already folded correctly.

28. Parathyroid hormone raises blood calcium concentration. Parathyroid hormone does all of the following EXCEPT:

- A)** stimulate the release of calcitonin from the thyroid gland
- B)** increase reabsorption of calcium in the kidney
- C)** stimulate osteoclast-mediated bone resorption
- D)** increase phosphate excretion in the urine
- E)** promote activation of vitamin D

Answer: A

PTH raises blood calcium by stimulating osteoclast-mediated bone resorption, increasing renal calcium reabsorption, promoting vitamin D activation (which also raises intestinal absorption) and increasing urinary phosphate excretion. Triggering calcitonin release is the exception: calcitonin is a separate thyroid hormone that lowers blood calcium, so attributing it to PTH reverses the two hormones' opposite effects.

29. An intravenous drip delivers 0.9% saline, whose osmolarity matches that of blood plasma. Red blood cells suspended in this solution will:

- A)** shrink because the solution is hypotonic
- B)** swell because the solution is hypertonic
- C)** shrink and crenate because the solution is hypertonic
- D)** swell and possibly lyse because the solution is hypotonic
- E)** keep their normal shape because the solution is isotonic

Answer: E

0.9% saline has the same osmolarity as plasma, making it isotonic: water moves into and out of the red cells at equal rates and they keep their normal shape. Predicting swelling (hypotonic) or shrinkage (hypertonic) contradicts the stated equality of osmolarity.

30. A catastrophe reduces a large population to a few randomly surviving individuals, which then rebuild the population. The restored population shows greatly reduced genetic diversity. This is an example of:

- A)** sexual selection
- B)** the bottleneck effect
- C)** the founder effect
- D)** gene flow
- E)** stabilising selection

Answer: B

A crash to a few randomly surviving individuals that then rebuild the population is the bottleneck effect, genetic drift that strips allelic diversity. The founder effect instead describes a few individuals leaving to colonise new habitat and start a new population elsewhere; here the population rebounds in place, so the bottleneck fits.

Section D - Quantitative Chemistry (Q31-44)

31. Sulfur forms sulfuric acid, H_2SO_4 . Which of the following correctly gives both the oxidation number of sulfur in H_2SO_4 and the position of sulfur in the periodic table?

- A)** -2 ; Period 4, Group 16
- B)** $+2$; Period 2, Group 16
- C)** $+4$; Period 3, Group 16
- D)** $+6$; Period 3, Group 17
- E)** $+6$; Period 3, Group 16

Answer: E

Step 1: oxidation numbers in H_2SO_4 sum to 0, with H at $+1$ and O at -2 , so $2(+1) + \text{S} + 4(-2) = 0$.

Step 2: $\text{S} = 8 - 2 = +6$

Step 3: sulfur sits in Period 3, Group 16, directly below oxygen; assigning it to Group 17 confuses it with chlorine, the halogen of that row.

32. What is the oxidation number of manganese in KMnO_4 ?

- A)** $+2$ **B)** $+4$ **C)** $+6$ **D)** $+7$ **E)** $+8$

Answer: D

In KMnO_4 potassium is $+1$ and oxygen is -2 , so $(+1) + \text{Mn} + 4(-2) = 0$ and $\text{Mn} = +7$

The $+8$ answer omits potassium's $+1$ from the sum, and $+2$ is the state of the Mn^{2+} ion produced when permanganate acts as an oxidising agent, not its state inside the ion.

33. Zinc metal is placed in copper(II) sulfate solution: $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$. Which statement about this reaction is correct?

- A)** Cu metal acts as the reducing agent
- B)** Zn^{2+} is reduced because it loses electrons
- C)** Cu^{2+} is reduced because it gains electrons
- D)** Zn is reduced because it gains electrons
- E)** Cu^{2+} is oxidised because it loses electrons

Answer: C

Zinc goes from 0 to $+2$ in ZnSO_4 , losing electrons, so zinc is oxidised and acts as the reducing agent. Copper goes from $+2$ in Cu^{2+} to 0 as metal, gaining those electrons, so Cu^{2+} is reduced. Pairing 'loses electrons' with 'reduced' is self-contradictory, and copper metal is the product, not the reducing agent.

34. What is the hydrogen ion concentration in a solution of pH 4.0?

- A) $1.0 \times 10^{-10} \text{ mol dm}^{-3}$
- B) $1.0 \times 10^{-3} \text{ mol dm}^{-3}$
- C) $1.0 \times 10^4 \text{ mol dm}^{-3}$
- D) $1.0 \times 10^{-4} \text{ mol dm}^{-3}$
- E) $4.0 \times 10^{-1} \text{ mol dm}^{-3}$

Answer: D

pH = $-\log[\text{H}^+]$, so pH 4.0 gives $[\text{H}^+] = 10^{-4} = 1.0 \times 10^{-4} \text{ mol dm}^{-3}$

The value $1.0 \times 10^{-10} \text{ mol dm}^{-3}$ is the hydroxide concentration instead, since $\text{pOH} = 14 - 4 = 10$, and 10^{-3} would correspond to pH 3.

35. At 25 °C, $K_w = 1.0 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$. A solution has $[\text{OH}^-] = 1.0 \times 10^{-4} \text{ mol dm}^{-3}$. What is the pH of this solution?

- A) 4 B) 7 C) 10 D) 11 E) 14

Answer: C

Step 1: $\text{pOH} = -\log(1.0 \times 10^{-4}) = 4$

Step 2: at 25 °C, $\text{pH} + \text{pOH} = 14$, so $\text{pH} = 14 - 4 = 10$

Reporting 4 quotes the pOH, and 7 would need a neutral solution with $[\text{OH}^-] = 10^{-7} \text{ mol dm}^{-3}$.

36. What is the pH of a $0.001 \text{ mol dm}^{-3}$ solution of hydrochloric acid, a strong monoprotic acid?

- A) -3 B) 0.001 C) 2 D) 3 E) 11

Answer: D

HCl is a strong monoprotic acid and dissociates completely, so $[\text{H}^+] = 0.001 = 10^{-3} \text{ mol dm}^{-3}$ and $\text{pH} = -\log(10^{-3}) = 3$. The value 11 is $14 - 3$, which is the pOH of the solution, and -3 is the logarithm with the sign slipped.

37. How many molecules are present in 22.4 dm^3 of oxygen gas at STP? (Molar volume at STP = $22.4 \text{ dm}^3 \text{ mol}^{-1}$; Avogadro constant = $6.02 \times 10^{23} \text{ mol}^{-1}$)

- A) 1.204×10^{24} B) 1.35×10^{25} C) 3.01×10^{23} D) 6.02×10^{23} E) 6.02×10^{22}

Answer: D

Step 1: moles of $\text{O}_2 = 22.4 \text{ dm}^3 / 22.4 \text{ dm}^3 \text{ mol}^{-1} = 1 \text{ mol}$

Step 2: molecules = $1 \text{ mol} \times 6.02 \times 10^{23} \text{ mol}^{-1} = 6.02 \times 10^{23} \text{ molecules}$

Doubling to 1.204×10^{24} assumes 2 mol of gas, and multiplying Avogadro's number by 22.4 treats a molar volume as a mole count.

38. 0.5 mol of sodium hydroxide is dissolved in water and the solution is made up to 250 cm^3 . What is the concentration of the solution in mol dm^{-3} ?

- A) $0.125 \text{ mol dm}^{-3}$ B) 0.5 mol dm^{-3} C) 1.25 mol dm^{-3} D) 2.0 mol dm^{-3} E) 5.0 mol dm^{-3}

Answer: D

Step 1: $250 \text{ cm}^3 = 0.250 \text{ dm}^3$

Step 2: concentration = $0.5 \text{ mol} / 0.250 \text{ dm}^3 = 2.0 \text{ mol dm}^{-3}$

The 0.125 answer multiplies by the volume instead of dividing, and 0.5 mol dm^{-3} ignores that the solution was diluted up to 250 cm^3 .

39. Which law states that the volume of a gas is directly proportional to the number of moles of gas present, provided the temperature and pressure are constant?

- A)** Avogadro's law
- B)** Boyle's law
- C)** Charles's law
- D)** Gay-Lussac's law
- E)** Dalton's law of partial pressures

Answer: A

Volume proportional to number of moles at constant temperature and pressure is Avogadro's law: equal volumes of gas at the same T and P contain equal moles. Charles's law links volume to absolute temperature, Boyle's law links volume to pressure, and Dalton's law deals with partial pressures, so none of those describes a moles-volume relationship.

40. A gas occupies 9.0 dm^3 at 27°C and constant pressure. It is then cooled to -73°C , still at the same pressure. What volume does it now occupy?

- A)** -24.3 dm^3
- B)** 3.0 dm^3
- C)** 6.0 dm^3
- D)** 13.5 dm^3
- E)** 24.3 dm^3

Answer: C

Step 1: convert to kelvin: $T_1 = 27 + 273 = 300 \text{ K}$ and $T_2 = -73 + 273 = 200 \text{ K}$

Step 2: $V_2 = 9.0 \times (200/300) = 6.0 \text{ dm}^3$

Inserting the Celsius readings directly produces the negative volume, and cooling at constant pressure must shrink the gas, so expansion answers like 13.5 dm^3 are impossible.

41. A gas occupies 500 cm^3 at a pressure of 100 kPa , the temperature remaining constant. The pressure is increased to 250 kPa . What is the new volume?

- A)** 50 cm^3
- B)** 100 cm^3
- C)** 200 cm^3
- D)** 250 cm^3
- E)** 1250 cm^3

Answer: C

Boyle's law at constant temperature: $P_1V_1 = P_2V_2$, so $V_2 = 500 \times (100/250) = 200 \text{ cm}^3$

Raising the pressure must compress the gas, so the 1250 cm^3 answer, which multiplies by the inverted ratio $250/100$, contradicts that, and 250 cm^3 would only fit a doubling of the pressure.

42. Each of the following compounds contains two carbon atoms. Which has the highest boiling point?

- A)** CH_3CH_3 (ethane)
- B)** CH_3COOH (ethanoic acid)
- C)** CH_3OCH_3 (methoxymethane)
- D)** CH_3CHO (ethanal)
- E)** $\text{CH}_3\text{CH}_2\text{OH}$ (ethanol)

Answer: B

Boiling point tracks intermolecular force strength. Ethanoic acid molecules form cyclic dimers held by two hydrogen bonds at once, giving about 118°C , well above ethanol's one-to-one hydrogen bonding at about 78°C . Ethanal, methoxymethane and ethane lack an O-H hydrogen-bond donor, so dipole and then dispersion forces make them progressively lower-boiling.

43. Which one of the following molecules is polar?

- A) NH_3 B) BF_3 C) CO_2 D) CCl_4 E) BeCl_2

Answer: A

NH_3 is pyramidal with a lone pair on nitrogen, so its three N-H bond dipoles point asymmetrically and do not cancel: it is polar. CO_2 and BeCl_2 (linear), BF_3 (trigonal planar) and CCl_4 (tetrahedral) are all symmetric, so their bond dipoles cancel and the molecules are non-polar despite having polar bonds.

44. Given the following data: $\text{C(s)} + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$, $\Delta H = -394 \text{ kJ mol}^{-1}$; $\text{CO(g)} + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$, $\Delta H = -283 \text{ kJ mol}^{-1}$. What is ΔH for the reaction $\text{C(s)} + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{CO(g)}$?

- A) -677 kJ mol^{-1} B) -394 kJ mol^{-1} C) -283 kJ mol^{-1} D) -111 kJ mol^{-1} E) $+111 \text{ kJ mol}^{-1}$

Answer: D

Step 1: reverse the second equation to $\text{CO}_2 \rightarrow \text{CO} + \frac{1}{2} \text{O}_2$, flipping its enthalpy change to $+283 \text{ kJ mol}^{-1}$. Step 2: add it to $\text{C(s)} + \text{O}_2 \rightarrow \text{CO}_2$ (enthalpy -394 kJ mol^{-1}); O_2 and CO_2 cancel, leaving $\text{C(s)} + \frac{1}{2} \text{O}_2 \rightarrow \text{CO}$.

Step 3: the total is $-394 + 283 = -111 \text{ kJ mol}^{-1}$

The -677 figure adds both given values without reversing the second equation, forgetting the sign flip.

Section E - Mathematics (Q45-53)

45. Solve for x : $\sqrt{3x - 2} = 4$.

- A) $x = 18$ B) $x = 14/3$ C) $x = 4$ D) $x = 2$ E) $x = 6$

Answer: E

Step 1: square both sides: $3x - 2 = 16$

Step 2: add 2 and divide by 3: $x = 18/3 = 6$

Step 3: check: the square root of $(18 - 2)$ is 4, so it is valid. The $14/3$ result comes from writing $3x = 16 - 2$, subtracting the 2 after squaring instead of adding it.

46. Solve for x : $\sqrt{x + 6} = x$. Which answer lists ALL valid solutions?

- A) $x = 6$
B) $x = 3$ only
C) $x = -2$ only
D) $x = 3$ and $x = -2$
E) $x = 2$

Answer: B

Step 1: square both sides: $x + 6 = x^2$, so $x^2 - x - 6 = 0$

Step 2: factorise: $(x - 3)(x + 2) = 0$, giving candidates $x = 3$ and $x = -2$

Step 3: substitute back: the square root of 9 is 3, which works, but the square root of 4 is 2, never -2, so -2 is extraneous and $x = 3$ is the only valid solution. Listing both candidates skips this essential check.

47. Solve the inequality: $5 - 3x \geq 20$.

- A) $x \geq -5$ B) $x \leq -5$ C) $x \geq 5$ D) $x < -5$ E) $x \leq 5$

Answer: B

Step 1: subtract 5: $-3x \geq 15$

Step 2: divide both sides by -3, which flips the inequality sign: $x \leq -5$. Keeping the direction ($x \geq -5$) is the classic sign-flip error, and using a strict $<$ loses the inclusive boundary of the original $\geq$.

48. Solve the inequality: $-2x + 7 > 15$.

- A) $x < -4$ B) $x < 4$ C) $x > -4$ D) $x > 4$ E) $x < -11$

Answer: A

Step 1: subtract 7: $-2x > 8$. Step 2: divide by -2 and flip the sign: $x < -4$. The answer $x > -4$ forgets the flip, and $x < -11$ comes from adding 7 to the right side (giving $-2x > 22$) instead of subtracting it.

49. If $f(x) = 3x^2 - 4x + 5$, what is $f(2)$?

- A) 4 B) 9 C) 13 D) 25 E) 33

Answer: B

Substitute $x = 2$: $f(2) = 3(2)^2 - 4(2) + 5 = 12 - 8 + 5 = 9$

The value 13 treats $-4x$ as -4 after substituting, and 33 comes from squaring the whole coefficient term as $(3x)^2 = 36$ instead of $3x^2 = 12$.

50. A 15 m ladder leans against a vertical wall with its foot 9 m from the base of the wall. How high up the wall does the ladder reach?

- A) 6 m B) 12 m C) 13 m D) 17.5 m E) 24 m

Answer: B

The ladder is the hypotenuse, so $\text{height}^2 = 15^2 - 9^2 = 225 - 81 = 144$ and the height is 12 m (a scaled 3-4-5 triple). Adding the squares, the square root of $(225 + 81)$ is about 17.5 m, which would make the wall the hypotenuse and contradicts the ladder leaning against it.

51. A rectangle has a width of 16 cm and a diagonal of 34 cm. What is its perimeter?

- A) 46 cm B) 62 cm C) 92 cm D) 100 cm E) 128 cm

Answer: C

Step 1: the diagonal is the hypotenuse, so the missing side = square root of $(34^2 - 16^2) = \text{square root of } (1156 - 256) = 30$ cm (a doubled 8-15-17 triple).

Step 2: perimeter = $2 \times (16 + 30) = 92$ cm

Using the diagonal as if it were a side gives $2 \times (16 + 34) = 100$ cm, and forgetting to double $(16 + 30)$ gives 46 cm.

52. Solve for x : $4(x - 2) = 2x + 2$.

- A) $x = 4$ B) $x = -3$ C) $x = 2$ D) $x = 10$ E) $x = 5$

Answer: E

Step 1: expand both terms: $4x - 8 = 2x + 2$

Step 2: $2x = 10$, so $x = 5$

Step 3: check $4(5 - 2) = 12$ and $2(5) + 2 = 12$

The answer $x = 2$ comes from distributing the 4 to only one term inside the bracket (writing $4x - 2$), and $x = 4$ from an arithmetic slip in the expansion.

53. Using $\pi = 3.14$, what is the area of a circle with a diameter of 12 cm?

- A) 37.68 cm² B) 75.36 cm² C) 113.04 cm² D) 226.08 cm² E) 452.16 cm²

Answer: C

Radius = $12 / 2 = 6$ cm, so area = $3.14 \times 6^2 = 3.14 \times 36 = 113.04$ cm²

The 452.16 cm² answer forgets to halve the diameter (3.14×144), and 37.68 cm² is the circumference, $2 \times 3.14 \times 6$, not the area.

Section F - Physics (Q54-62)

54. A car travels at 72 km/h. What is this speed in m/s?

- A) 7.2 m/s B) 20 m/s C) 24 m/s D) 36 m/s E) 240 m/s

Answer: B

Convert with the fixed factor 3.6: $72 \text{ km/h} / 3.6 = 20 \text{ m/s}$, since 1 km/h equals 1000 m per 3600 s. The m/s number must be smaller than the km/h number, so 36 or 240 m/s, which come from dividing by 3 or from multiplying, are wrong.

55. A cyclist rides at 15 m/s. What is this speed in km/h?

- A) 4.2 km/h B) 5 km/h C) 45 km/h D) 54 km/h E) 150 km/h

Answer: D

Multiply by 3.6: $15 \text{ m/s} \times 3.6 = 54 \text{ km/h}$ ($15 \times 3 = 45$ plus $15 \times 0.6 = 9$), because 1 m/s covers 3.6 km in an hour. The converted value must be bigger, so 4.2 or 5 km/h, produced by dividing instead of multiplying, cannot be right.

56. Two 6 Ω resistors are connected in parallel. What is the total resistance?

- A) $1/3 \Omega$ B) 3 Ω C) 6 Ω D) 12 Ω E) 36 Ω

Answer: B

For parallel resistors, $1/R = 1/6 + 1/6 = 1/3$, so $R = 3 \text{ ohm}$ (two equal resistors in parallel simply halve). A parallel total is always smaller than the smallest branch, so 6 ohm or the series-style 12 ohm are impossible, and leaving $1/3 \text{ ohm}$ forgets the final inversion of $1/R$.

57. Three resistors of 4 Ω , 4 Ω and 2 Ω are connected in parallel. What is the total resistance?

- A) 0.25 Ω B) 1 Ω C) 2 Ω D) 3.3 Ω E) 10 Ω

Answer: B

Step 1: $1/R = 1/4 + 1/4 + 1/2 = 0.25 + 0.25 + 0.5 = 1$

Step 2: invert: $R = 1 \text{ ohm}$

A parallel total must come out below the smallest branch (2 ohm), so the plain sum of 10 ohm would only be the series result, not this connection's.

58. A current of 3 A flows through a 4 Ω resistor. What power is dissipated?

- A) 7 W B) 12 W C) 36 W D) 48 W E) 144 W

Answer: C

$P = I^2 R = 3^2 \times 4 = 9 \times 4 = 36 \text{ W}$

The 12 W answer uses $P = IR$ without squaring the current, 144 W squares the product IR instead, and 48 W squares the resistance; in this formula only the current is squared.

59. An appliance draws a current of 2.5 A when connected to a 12 V supply. What power does it use?

- A) 0.21 W B) 4.8 W C) 14.5 W D) 30 W E) 60 W

Answer: D

$$P = VI = 12 \text{ V} \times 2.5 \text{ A} = 30 \text{ W}$$

The 4.8 W answer divides voltage by current and 14.5 W adds them; electrical power is the product of voltage and current, never their quotient or sum.

60. What is the gauge pressure at a depth of 5 m in water? ($\rho = 1000 \text{ kg/m}^3$, $g = 9.8 \text{ m/s}^2$)

- A) 4,900 Pa B) 9,800 Pa C) 49,000 Pa D) 49 Pa E) 490,000 Pa

Answer: C

$$\text{Gauge pressure} = \text{density} \times g \times \text{depth} = 1000 \times 9.8 \times 5 = 49,000 \text{ Pa}$$

The 4,900 Pa answer drops a factor of ten, and 9,800 Pa equals just density times g with the depth left out; all three factors multiply in this relation.

61. A block has a mass of 540 g and a volume of 200 cm^3 . What is its density, and which metal is it consistent with?

- A) 0.37 g/cm^3 — a light alloy
B) 2.7 g/cm^3 — aluminium
C) 27 g/cm^3 — a heavy alloy
D) 108 g/cm^3 — gold
E) 740 g/cm^3 — lead

Answer: B

Density = mass / volume = $540 \text{ g} / 200 \text{ cm}^3 = 2.7 \text{ g/cm}^3$, which matches aluminium. The 0.37 g/cm^3 answer inverts the division ($200/540$); mass divided by a smaller volume must exceed 1, and no real metal approaches 740 g/cm^3 .

62. A lamp produces a light intensity of 90 W/m^2 at a distance of 1 m. What is the intensity at 3 m from the lamp?

- A) 10 W/m^2 B) 30 W/m^2 C) 90 W/m^2 D) 270 W/m^2 E) 810 W/m^2

Answer: A

Intensity follows the inverse-square law: intensity is proportional to $1/\text{distance}^2$. Tripling the distance from 1 m to 3 m divides the intensity by $3^2 = 9$, so $90 / 9 = 10 \text{ W/m}^2$. The 30 W/m^2 answer divides by 3 only, treating the law as inverse rather than inverse-square.

Section G - Logic & Reasoning (Q63-70)

63. 'If Maria catches the 8:00 train, she arrives before noon.' Maria did not arrive before noon. Which conclusion necessarily follows?

- A) If Maria does not catch the 8:00 train, she does not arrive before noon.
- B) If Maria arrives before noon, she caught the 8:00 train.
- C) Maria missed the 8:00 train because she overslept.
- D) Maria caught the 8:00 train, but the train was delayed.
- E) Maria did not catch the 8:00 train.

Answer: E

The statement is a conditional: catching the 8:00 train guarantees arriving before noon. From 'she did not arrive before noon', the valid inference is the contrapositive: she did not catch the 8:00 train. The delay and oversleeping stories add facts that contradict this conclusion, and neither the converse ('arrived before noon, so she caught the train') nor the inverse ('no train, so no arrival') is licensed by the original.

64. A technician states: 'If the sample is contaminated, the result is unreliable.' Which statement, if true, contradicts this?

- A) The sample is contaminated and the result is reliable.
- B) The sample is not contaminated and the result is unreliable.
- C) The sample is contaminated and the result is unreliable.
- D) If the sample is not contaminated, the result is reliable.
- E) The sample is not contaminated and the result is reliable.

Answer: A

A conditional is false only when its antecedent is true and its consequent is false: the sample is contaminated and yet the result is reliable. Every other combination, including an uncontaminated sample with an unreliable result, leaves the statement intact, because the promise covers only contaminated samples.

65. Anna reasons: 'If the alarm rings, the neighbours call the police. The alarm did not ring, so the neighbours did not call the police.' Her reasoning is...

- A) invalid: the original conditional itself must be false because the neighbours might call anyway
- B) valid: it correctly applies the contrapositive of the original statement
- C) valid: the inverse of a conditional always follows from it
- D) valid: the neighbours can only call the police when the alarm rings
- E) invalid: it denies the antecedent, and the inverse is not equivalent to the original conditional

Answer: E

The inference runs from 'the alarm did not ring' to 'the neighbours did not call', which is denying the antecedent. The neighbours could still call for some other reason, so the conclusion does not follow; the inverse of a conditional is not equivalent to it. The valid pattern would be the contrapositive: no call from the neighbours implies the alarm did not ring.

66. All cardiologists are doctors. Some teachers are doctors. No cardiologist is a teacher. Which conclusion follows necessarily?

- A)** All teachers are doctors.
- B)** All doctors are either cardiologists or teachers.
- C)** No teacher is a doctor.
- D)** Some cardiologists are teachers.
- E)** Some teachers are not cardiologists.

Answer: E

Premise two guarantees at least one person who is both a teacher and a doctor. Premise three bars every cardiologist from being a teacher, so that person is a teacher who is not a cardiologist: 'some teachers are not cardiologists' follows necessarily. 'All teachers are doctors' overreaches the premises, and 'some cardiologists are teachers' contradicts premise three outright.

67. No reptiles are warm-blooded. All snakes are reptiles. Which conclusion follows necessarily?

- A)** No snakes are warm-blooded.
- B)** All warm-blooded animals are reptiles.
- C)** Some snakes are warm-blooded.
- D)** No animal other than reptiles is warm-blooded.
- E)** Some reptiles are warm-blooded.

Answer: A

All snakes are reptiles and the reptile set is disjoint from the warm-blooded set, so every snake is cold-blooded: no snakes are warm-blooded. The other options either reverse the inclusion ('all warm-blooded animals are reptiles') or assert warm-blooded reptiles, which the premises exclude.

68. A study of 5,000 cyclists found that riders without helmets were more than three times as likely to sustain head injuries as riders wearing helmets. Which conclusion is best supported?

- A)** In this study, unhelmeted riders had a higher incidence of head injury than helmeted riders.
- B)** Cycling without a helmet causes riders to crash more often.
- C)** Unhelmeted riders take more risks than helmeted riders.
- D)** Helmets prevent all head injuries in cycling accidents.
- E)** Helmet laws should be introduced to reduce all cycling injuries.

Answer: A

The data license only the within-study comparison: unhelmeted riders had a higher incidence of head injury than helmeted riders. Risk-taking behaviour, crash frequency, complete prevention and legislative recommendations all import causal or universal claims that observational numbers never established.

69. A survey of 2,000 employees found that those who walk to work reported, on average, higher job satisfaction than those who commute by car. Which conclusion follows?

- A)** Car commuters would become more satisfied if they switched to walking.
- B)** Employers should subsidise walking programmes in order to raise productivity.
- C)** Walking to work causes greater job satisfaction.
- D)** Every employee who walks to work is more satisfied than every employee who drives.
- E)** Among surveyed employees, walking commuters reported higher average satisfaction than car commuters.

Answer: E

The survey supports exactly one comparison: among surveyed employees, walking commuters reported higher average satisfaction than car commuters. An average hides individual variation, so 'every walker more satisfied than every driver' overreaches, and the causal claim, the switching counterfactual and the productivity recommendation all go beyond the correlational data.

70. What is the angle between the hour hand and the minute hand of an analogue clock at 3:15?

A) 0 degrees B) 7.5 degrees C) 15 degrees D) 30 degrees E) 82.5 degrees

Answer: B

Step 1: the minute hand at 15 minutes is $15 \times 6 = 90$ degrees from 12. Step 2: the hour hand moves 30 degrees per hour plus 0.5 degrees per minute, so it is $3 \times 30 + 15 \times 0.5 = 97.5$ degrees from 12. Step 3: the angle between them is $97.5 - 90 = 7.5$ degrees. The zero-degree claim assumes both hands sit on the 3, ignoring the hour hand's quarter-hour drift toward 4.